Data Handling: Import, Cleaning and Visualisation

Exercise 4: Data Import and Data Preparation/Manipulation

Ulrich Matter

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Exercise A

Read Section 11.-11.4 in https://r4ds.had.co.nz/data-import.html and solve the problem posed in Section 11.4.2 ("Problems") by going step-by-step through the instructions. Hint: You will need to have the `readr` package installed and loaded (should be also installed/loaded with `tidyverse`).

Exercise B

1. Download the file `tv_ownership.dat` posted on Canvas and save it in your `r_course/data` folder.
2. The file is an excerpt of a dataset provided by the US Federal Communications Commission (FCC) with detailed data on TV and Radio stations ownership. The excerpt is in the original format as provided by the FCC. Figure out how to read this dataset into R with the functions provided in the `tidyverse/readr`-packages.
3. Write your solution down in an R-Script and document in this script why certain problems occur and how you solved them (using comments).
4. Once you have properly imported the TV ownership data, prepare a text file for upload on GitHub classroom. Namely, export the value in the 7th column for the 15th observation to a text file named "tv_ownership_value.txt" (save the file in your `data` folder).
5. Upload "tv_ownership_value.txt" to GitHub classroom.

Exercise C

1. Visit the EU Open Data Portal on COVID cases. (Do you remember this website from Exercise Session 2?)
2. Download the "COVID-19 Cases Worldwide" data in JSON format. However, do not download these data manually (i.e., do not click-and-save). Instead, load the data directly into RStudio from the web. Hint 1: consider the `jsonlite` package. Hint2: the direct link to the JSON file (https://opendata.ecdc.europa.eu/covid19/casedistribution/json/) may be helpful.
3. Reformat the JSON data that you have imported into RStudio to a rectangular format (such that each record is represented as a row and each field as a column).

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. `for`-loops are typically used in a situation where a sequence of commands needs to be executed for a well-known number of times.

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Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct response statements need to be marked!

1. After executing the follwing lines of R code, which value is stored in `a[3]`?

```
a <- c(1:3)
```

```
a <-  a*3
```

A) `3`
B) `1:3`
C) `9`
D) `27`

Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. The following R code was executed on the command line and the corresponding error occurred.

```
read_csv("data/file.csv")
```

```
Error: 'data/file.csv' does not exist in current working directory ('/home/user/my_projects').
```

Which of the following statements is correct in this context (assuming `file.csv` actually is stored somewhere on this computer‘s harddrive)?

A) `read_csv()` fails to parse `file.csv` because the wrong column delimiter is used (commas).
B) `read_csv()` is not the right R function to import `file.csv`.
C) Given the current working directory of this R session, the relative path “data/file.csv” is wrong and `read_csv()` therefore does not find the file.
D) `file.csv` is generally not a valid file name.